

ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \quad (1.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.1b)$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \epsilon \frac{\partial E_x}{\partial t} + \sigma E_x \quad (1.2a)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \epsilon \frac{\partial E_y}{\partial t} + \sigma E_y \quad (1.2b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \epsilon \frac{\partial E_z}{\partial t} + \sigma E_z \quad (1.2c)$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \quad (1.3a)$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \quad (1.3b)$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \quad (1.3c)$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i . The implementation uses

$\mathbf{E}_\alpha$: cell-centered in the α direction and nodal in the other two directions,

$\mathbf{H}_\alpha$: nodal in the α direction and cell-centered in the other two directions.

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i + \frac{1}{2}$. They are encoded in the **MultiFab** staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j + \frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j + 1$. Therefore

$$\left. \frac{\partial E_z}{\partial y} \right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}.$$

Similarly, $E_{x,i,j,k}$ is located at j , while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j - \frac{1}{2}$ and $j + \frac{1}{2}$, so

$$\left. \frac{\partial H_z}{\partial y} \right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}.$$

This is why the magnetic update below uses forward differences such as $j + 1$ and $k + 1$, while the electric update uses backward differences such as $j - 1$ and $k - 1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update **H** by a half step, then update **E** by a full step, and finally update **H** by another half step. Define the update coefficients

$$C_a = \frac{1 - \frac{\sigma \Delta t}{2\epsilon}}{1 + \frac{\sigma \Delta t}{2\epsilon}}, \quad C_b = \frac{\frac{\Delta t}{\epsilon}}{1 + \frac{\sigma \Delta t}{2\epsilon}}, \quad D_b = \frac{\Delta t}{\mu}, \quad (2.1)$$

where Δt is the time step, ϵ is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \frac{D_b}{2} \left[\frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta z} \right] \quad (2.2a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \frac{D_b}{2} \left[\frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta x} \right] \quad (2.2b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - \frac{D_b}{2} \left[\frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta y} \right]. \quad (2.2c)$$

In the AMReX Yee grid, the magnetic field updates are implemented as

$$B_x(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.3a)$$

$$B_y(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.3b)$$

$$B_z(i, j, k) = \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.3c)$$

2.2 Electric Full-Step

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_a E_{x,i+\frac{1}{2},j,k}^n + C_b \left[\frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.4a)$$

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_a E_{y,i,j+\frac{1}{2},k}^n + C_b \left[\frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_a E_{z,i,j,k+\frac{1}{2}}^n + C_b \left[\frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.4c)$$

In the AMReX Yee grid, the electric field updates are implemented as

$$E_x(i, j, k) + = c^2 \Delta t \left[\frac{B_z(i, j, k) - B_z(i, j - 1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k - 1)}{\Delta z} \right] \quad (2.5a)$$

$$E_y(i, j, k) + = c^2 \Delta t \left[\frac{B_x(i, j, k) - B_x(i, j, k - 1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i - 1, j, k)}{\Delta x} \right] \quad (2.5b)$$

$$E_z(i, j, k) + = c^2 \Delta t \left[\frac{B_y(i, j, k) - B_y(i - 1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j - 1, k)}{\Delta y} \right]. \quad (2.5c)$$

2.3 Second Magnetic Half-Step

$$H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.6a)$$

$$H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.6b)$$

$$H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.6c)$$

In the AMReX Yee grid, the second magnetic field updates are implemented as

$$B_x(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.7a)$$

$$B_y(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.7b)$$

$$B_z(i, j, k) - = \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.7c)$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \quad c = \frac{1}{\sqrt{\mu\epsilon}}. \quad (2.8)$$

3 ADI Method

3.1 First ADI Half-Step

By equation (1.2a),

$$\epsilon_{i+\frac{1}{2}, j+\frac{1}{2}, k} \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2}, j+\frac{1}{2}, k} \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} + E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^n}{2} = [H_1], \quad (3.1)$$

where

$$[H_1] = \frac{H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^n - H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k-\frac{1}{2}}^n}{\Delta z}. \quad (3.2)$$

Rearranging gives

$$\left(\frac{2\epsilon}{\Delta t} + \frac{\sigma}{2} \right) E_x^{n+\frac{1}{2}} = \left(\frac{2\epsilon}{\Delta t} - \frac{\sigma}{2} \right) E_x^n + [H_1]. \quad (3.3)$$

Hence

$$E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} = C_{a, i+\frac{1}{2}, j+\frac{1}{2}, k} E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k}^n + C_{b, i+\frac{1}{2}, j+\frac{1}{2}, k} [H_1], \quad (3.4)$$

where

$$C_a = \frac{4\epsilon - \sigma\Delta t}{4\epsilon + \sigma\Delta t}, \quad (3.5)$$

and

$$C_b = \frac{2\Delta t}{4\epsilon + \sigma\Delta t}. \quad (3.6)$$

By equation (1.3c),

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right), \quad (3.7)$$

where

$$D_{b,i+\frac{1}{2},j+\frac{1}{2},k} = \frac{\Delta t}{2\mu_{i+\frac{1}{2},j+\frac{1}{2},k}}. \quad (3.8)$$

Substituting the magnetic update into (3.4) and eliminate $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left[H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n \right. \\ &+ D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} \right. \\ &\quad \left. \left. - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right) \right. \\ &- H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \\ &- D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}}}{\Delta y} \right. \\ &\quad \left. \left. - \frac{E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n}{\Delta x} \right) \right] \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right). \end{aligned} \quad (3.9)$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
& -\frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} \\
& + \left[1 - \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\
& - \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} \\
& = C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n \\
& + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
& - \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y \Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\
& + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y \Delta x} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
& - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right).
\end{aligned}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} & = p_{x1} E_{x,i+\frac{1}{2},j,k}^n \\
& + \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
& - \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
& + \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
& - \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right). \tag{3.10}
\end{aligned}$$

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$